

Fake News Detection: Project Report

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May 2025

1. Introduction

1.1 Problem Statement

In today's digital era, the rapid spread of misinformation poses a significant threat to informed public discourse and democratic processes. Fake news articles can manipulate public opinion, influence political outcomes, and undermine trust in legitimate media. This project addresses this challenge by developing an automated system for detecting fake news using machine learning techniques.

1.2 Objectives

- To develop a machine learning-based system capable of distinguishing between fake and real news articles
- To create and compare multiple classification models for fake news detection
- To implement a user-friendly web application for real-time fake news detection
- To achieve high accuracy while maintaining computational efficiency

1.3 Scope

This project focuses on text-based fake news detection using natural language processing and machine learning classification techniques. The system analyzes the content of news articles to determine their credibility based on patterns learned from a labeled dataset of real and fake news.

2. Related Work

Several approaches to automated fake news detection have been proposed in recent years:

2.1 Content-Based Approaches

Content-based methods focus on extracting linguistic features from the text itself:

- Pérez-Rosas et al. (2018) used linguistic features including n-grams, readability, and syntax to classify fake news with 76% accuracy
- Potthast et al. (2018) applied stylometric analysis to identify writing style patterns associated with fake news

2.2 Machine Learning Approaches

Recent work has explored various machine learning techniques:

- Ahmed et al. (2017) used n-gram analysis and various machine learning algorithms, finding that TF-IDF with Linear SVC achieved 92% accuracy
- Singhanian et al. (2017) applied deep learning models including CNNs, achieving accuracy rates of up to 95.7%

3. Methodology

3.1 Dataset Description

The project uses Kaggle dataset containing thousands of labeled news articles. Each entry includes:

- News article title

- Text content
- Label (REAL or FAKE)

The dataset is well-balanced with approximately equal numbers of real and fake news articles, eliminating potential bias in model training.

3.2 Data Preprocessing

3.2.1 Handling Missing Values

We identified and addressed null values in the dataset to ensure data integrity:

- Articles with missing titles were populated with placeholder values
- Articles with missing text content were removed as they provided insufficient information for analysis

3.2.2 Text Cleaning

Raw text data underwent several preprocessing steps:

- Conversion to lowercase to maintain consistency
- Removal of special characters and punctuation
- Elimination of stopwords (common words with limited discriminative value)
- Stemming to reduce words to their root forms
- Removal of HTML tags and URLs

3.2.3 Feature Extraction

We employed Term Frequency-Inverse Document Frequency (TF-IDF) vectorization to convert the processed text into a numerical format suitable for machine learning algorithms. This approach:

- Captures the importance of words in each document
- Downweights terms that appear frequently across the corpus
- Creates a sparse matrix representation of features

3.3 Model Implementation

3.3.1 Logistic Regression

Logistic Regression was implemented as a baseline model due to its:

- Simplicity and interpretability
- Effectiveness for text classification tasks
- Computational efficiency
- Strong performance with high-dimensional data

We optimized the model using L2 regularization to prevent overfitting.

3.3.2 Random Forest

Random Forest was implemented as an ensemble method that:

- Combines multiple decision trees to improve accuracy and robustness
- Reduces overfitting through bagging and feature randomization
- Handles high-dimensional data effectively
- Provides feature importance measures

3.3.3 Model Training and Hyperparameter Tuning

Models were trained using a train-test split of 80:20. We employed:

- Cross-validation to ensure robust evaluation
- Early stopping to prevent overfitting

3.4 Deployment Architecture

The system was deployed as a web application with the following components:

- Pickle files to store the trained models and vectorizers
- HTML frontend for user interaction
- Deployment on Render cloud platform for scalability and accessibility

The application is deployed and accessible at:

<https://fake-news-detection-y6tw.onrender.com/>

4. Results and Evaluation

4.1 Performance Metrics

We evaluated our models using multiple metrics:

Metrics	Logistic Regression	Random Forest
Accuracy	98.2%	96.36%
Precision(real)	98%	98%
Precision(fake)	99%	95%
Recall(real)	98%	95%
Recall(fake)	98%	98%
F1-Score(real)	98%	96%
F1-Score(fake)	98%	97%
Cross-Validation	98.19%	97.19%

4.3 Feature Importance Analysis

Analysis of feature importance revealed significant indicators of fake news:

- Certain emotional and sensationalist terms had high discriminative power
- Linguistic patterns such as excessive use of capitalization and punctuation were strong predictors
- Specific names and topics showed correlation with fake news classification

4.4 Error Analysis

Common misclassifications occurred in:

- Satirical articles that mimic fake news but are labeled as legitimate content
- Opinion pieces with strong language similar to that found in fake news

- Articles discussing fake news (meta-reporting) being classified as fake news

5. Web Application

5.1 User Interface

The web application provides:

- A clean, intuitive interface for users to input news text
- Real-time classification
- Responsive design for both desktop and mobile users

5.2 Backend Architecture

The application employs:

- Efficient model loading and caching
- Text preprocessing identical to the training pipeline
- Optimized inference for quick response times

5.3 Deployment Details

The application is deployed on Render with:

- Continuous integration and deployment from the GitHub repository
- Reliable uptime and performance monitoring

6. Limitations and Future Work

6.1 Current Limitations

- The model relies solely on text content and doesn't consider external factors like source credibility

- Language limitations (currently optimized for English content only)
- Limited context understanding in very short texts
- Vulnerability to adversarial attacks that deliberately avoid detection patterns

6.2 Future Improvements

Potential enhancements include:

- Incorporating additional models like BERT or other transformers for improved performance
- Adding URL credibility checking and metadata analysis
- Implementing multi-language support
- Developing an explanation component to highlight suspicious parts of the text

7. Conclusion

This project successfully demonstrates the effectiveness of machine learning approaches in detecting fake news articles. By combining careful data preprocessing, feature engineering, and appropriate model selection, we achieved classification accuracy exceeding 98%, significantly outperforming many existing solutions.

The deployed web application provides a practical tool for users to verify news content, contributing to the broader effort to combat misinformation online. While limitations exist, the foundation established in this project provides a solid platform for future enhancements and extensions.

8. References

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